

Development of a software platform for target identification, drug target interaction, and pharmacokinetic properties prediction

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Abstract. The primary objective of designing this web-based platform is to provide a single, unified environment that integrates all three major stages of the computational drug repurposing/discovery workflow. We combine target identification, drug-target interaction analysis, and pharmacokinetics prediction into one platform. The system eliminates the need for multiple specialized tools and complex data handling procedures. The platform is specifically designed to be user-friendly and accessible to individuals without a formal background in biology or computer science. Complex analytical processes such as data preprocessing, feature extraction, and AI model execution are automated and presented through interactive graphical interfaces, drop-down menus, and predefined options. This design enables users to efficiently upload data, perform analyses, and interpret results without requiring advanced programming skills or biological expertise. Overall, the platform simplifies data handling, reduces technical barriers, and facilitates broader adoption of AI-driven approaches for Pancreatic Ductal Adenocarcinoma research.

Keywords. Software, Target, Drug, Drug- Target Integration, Pharmacokinetics

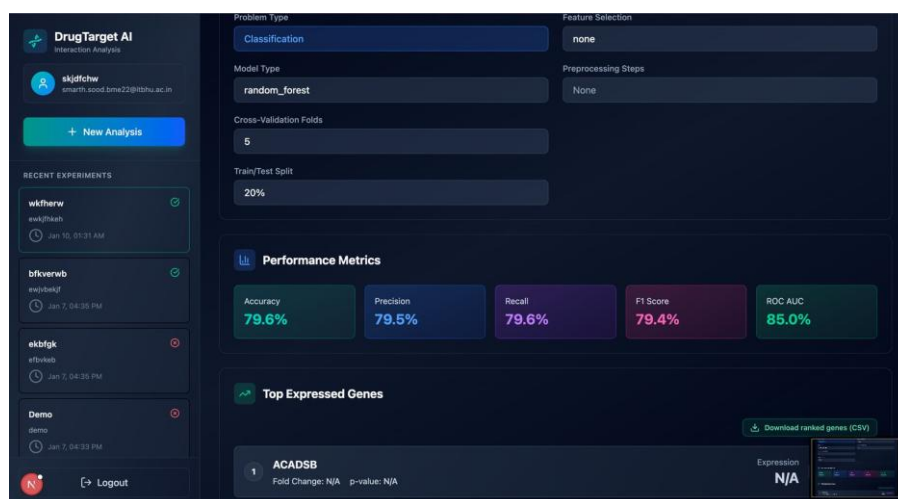


Figure 1. Web-based interface for the Proposed software platform.

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1. Introduction

Target identification

The first interface of the platform focuses on identifying potential therapeutic targets associated with PDAC. Raw gene expression data are provided as input to the system. Users can select appropriate preprocessing options through an interactive drop-down interface, including data normalization and filtering steps. Subsequently, AI-based algorithms and relevant training and testing parameters are configured within the platform, and the performance will be evaluated. Based on these selections, predictive models are trained to identify molecular targets that significantly differentiate between normal and disease conditions[1].

Drug library Preparation and Drug Target interaction

Following target identification, the platform enables drug library preparation through an integrated interface linked to the DepMap database. Candidate drugs corresponding to the identified targets are curated within the platform. Molecular descriptors and fingerprints are extracted for both targets and drug compounds. Using these features, AI-driven predictive models are developed to estimate drug-target binding energies and dissociation constants, facilitating the assessment of interaction strength and therapeutic potential[2].

Pharmacokinetics Prediction

In the final module, the platform evaluates the pharmacokinetic profiles of the prepared drug library. AI-based models are employed to predict ADMET (Absorption, Distribution, Metabolism, Excretion, and Toxicity) properties of the candidate compounds. This module assists in prioritizing drugs with favorable pharmacokinetic and safety characteristics for further experimental validation[3].

2. Method

The proposed software platform is designed as an integrated, end-to-end solution for cancer and genetic disorder research, leveraging Artificial Intelligence (AI) based methodologies. The platform consolidates three stages of the drug repurposing and evaluation pipeline into a single unified interface: (1) target identification, (2) drug-target interaction prediction, and (3) pharmacokinetics property prediction.

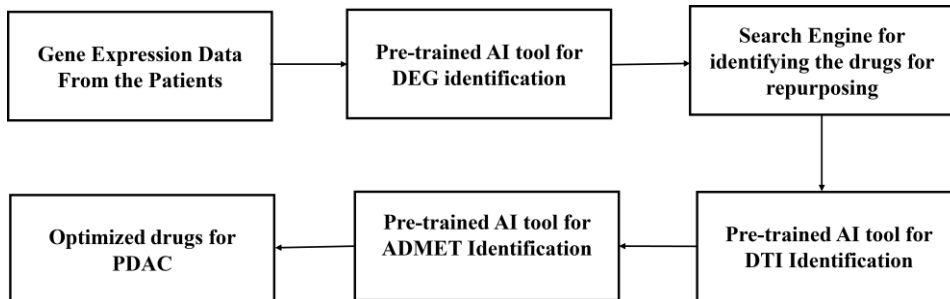


Figure 2. Process Pipeline of the study

3. Results

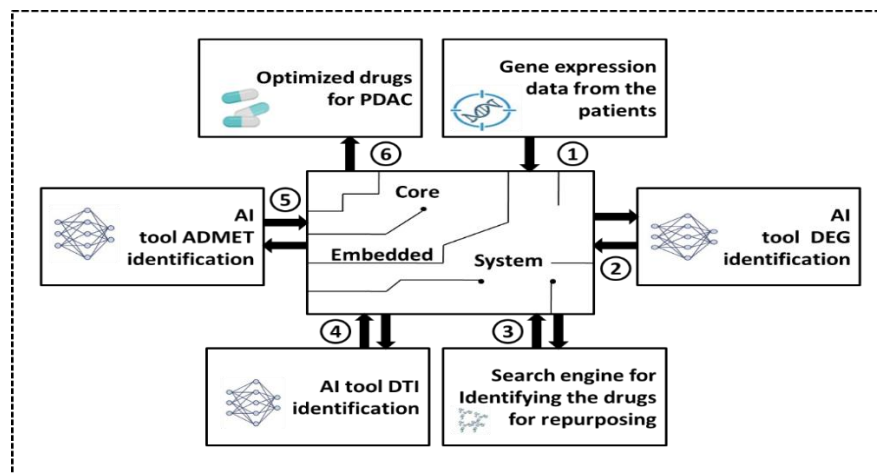


Figure 3. Smart Drug Repurposing Platform

References

- [1]Pragya, P., Govarthan, P. K., Nayak, M., Mukherjee, S., & Agastinose Ronickom, J. F. (2024). Establishment of three gene prognostic markers in pancreatic ductal adenocarcinoma using machine learning approach. *Journal of Medical and Biological Engineering*, 44(3), 357-365.
- [2]Pragya, Prince, A. A., & Ronickom, J. F. A. (2025). Binding Affinity Prediction for Pancreatic Ductal Adenocarcinoma Using Drug-Target Descriptors and Artificial Intelligence. *IEEE Access*.
- [3]Pragya, Ronickom, J. F. A. (2026). Artificial Intelligence Driven Virtual Screening and Molecular Docking Approaches Identified LIFR, BTG2, EPHX2, and PAK3 as Targets and BI-2536, AP-24534, and AZ-628 as Repurposed Drugs for PDAC. *IEEE Transactions on Computational Biology and Bioinformatics*.